

How to create a virtual drive for FLEX

 flexonsbd.blogspot.com/2018/04/flexvirtual-drive.html

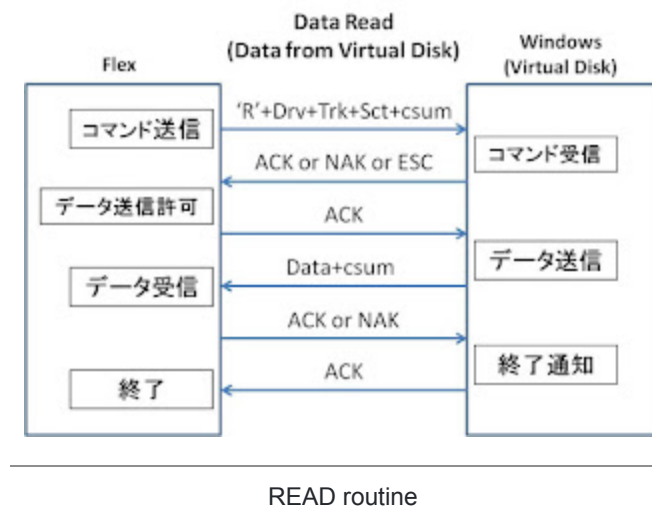
Saturday, April 21, 2018

How to create a Virtual Drive for FLEX

To create a virtual drive for FLEX, the following is required:

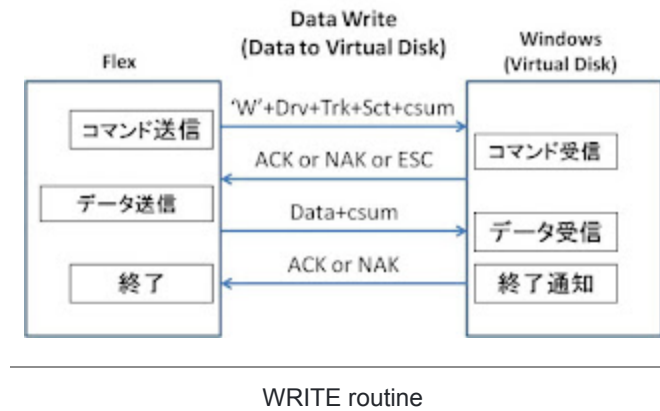
- (1) FLEX side: Modify the FLEX system's Disk Driver routine for use with the virtual driver.
- (2) Windows side: A program that receives commands from FLEX and reads and writes sector data accordingly.

(1) In the case of FLEX9 for 6809, the FLEX system's driver routine consists of nine functions: READ, WRITE, VERIFY, RESTORE, DRIVE, CHKRDY, QUICK, INIT, WARM, and SEEK, but the two most important are READ and WRITE. A conceptual diagram is shown below.



In short, by sending READ and WRITE command strings from FLEX, Windows will read sectors in the DSK file set in the Virtual Drive and return data, or record data to sectors accordingly.

Other routines can be handled solely by Windows.



(2) The Windows program basically only needs to read and write to the corresponding sector according to the drive number, track number, and sector number in the received command.

However, this alone is not practical as it does not allow for data or program exchange with Windows, so a separate program was created and used to transfer files to the virtual disk file on Windows using drag and drop, etc.

To create this, it was necessary to implement something almost identical to FLEX's FMS (File Management System), and it was created through trial and error, so it is not yet at a perfect stage, but it works for the time being. (In creating this, I referenced the idea of the X1 program CPMforX1DiskExplorer by " [Hissori X1](#)

". I am grateful.) This time, in creating a single-board computer, I created a Windows application that integrates these. That is [FlexDrvWin](#) , which I posted an image of in the blog on April 16th .

~~(I have just started blogging and do not yet know how to publish programs. I plan to publish them as soon as I figure it out.)~~ (Added on 5/3/2018: Published on GitHub.)